



Language-independent Profile-based Tag Recommendation for Community Question Answering Systems

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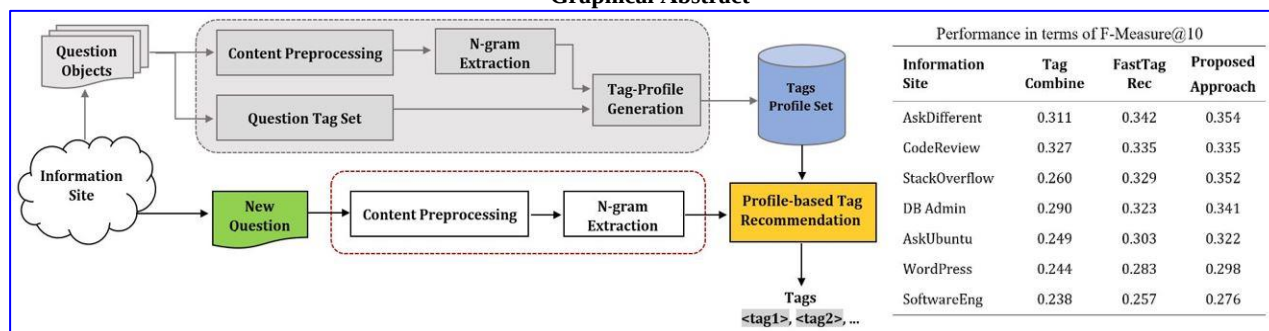
Similarity Measure

ABSTRACT

Community question-answering (CQA) systems are helpful for knowledge sharing. However, they can become difficult to manage as the number of questions and answers increases. Effective tag recommendation facilitates the discovery of relevant material, yet prevailing methods typically depend on language-specific resources or necessitate sophisticated Natural Language Processing (NLP) tools, rendering them unsuitable for less-resourced languages. This paper introduces a novel profile-based tag recommendation strategy that transcends language and structural barriers. The approach leverages raw text data without the need for complex text mining tools. By constructing distinct profiles for each tag from key terms in associated questions, the method enables a nuanced content association. An adaptation of the Term Frequency-Inverse Document Frequency (TF-IDF) metric is proposed to calculate similarity and recommend tags aligned with these profiles. The efficacy of this approach is validated across datasets in both English and Persian, showcasing comparable to or superior recall rates against baseline models and contemporary advanced systems. This methodology is straightforward to implement, offering a valuable tool for enhancing content accessibility in CQA platforms, particularly for low-resource languages.

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Graphical Abstract



1. INTRODUCTION

The rapid growth of information technology has spurred the development of community question-answering systems (CQA systems) (1). These systems provide user interfaces that empower users, including developers, to seek guidance from others when they encounter

challenges (2, 3). Within a CQA system, each unit of information is called a "Question Object." This object contains various components, including the question title, question body (also known as question details), and question tags. Figure 1 provides an example of a Question Object.

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A recommender module is a key building block for modern information site (4). The most important feature of this module is the use of question tags. As a prerequisite for asking a question, the questioner should assign relevant tags that accurately describe the question. These tags also help to organize the system content in a more efficient way (5). Several studies have investigated the impact of using tags in CQA systems and highlighted numerous advantages of this feature (6). Notably, it enables users to easily find related questions, learn various aspects of a field by selecting the relevant tag, and reduce duplicate questions. Furthermore, the inclusion of appropriate tags increases the chance that the question will be seen by domain experts and users with similar experiences.

However, while tags offer significant benefits, they also present challenges. As technology is constantly evolving, CQA systems cannot limit the terms that can be used for labeling. Additionally, different users may have different understandings of concepts and technologies, leading to differences in the terminology they use to refer to similar or related questions (7). This can lead to a number of problems, particularly the tag synonym problem.

The tag synonym problem occurs when users use different expressions for the same concept, such as variations in spelling, abbreviations, capitalization, separators, or spacing. For example, for the tag "command line", several synonyms have been added to the Tags collection, including "cmd", "cli", and "console". Figure 2 shows the tag synonym page of Ask Ubuntu which is manually provided by advanced users. This collection is minimal because manually collecting all tags is difficult and time-consuming.

The tag synonym problem makes it difficult to categorize questions. This is because related questions are no longer linked with the same tags, making

searching for and finding relevant content difficult (8). Over time, this problem can lead to tag redundancy (also known as tag explosion), which refers to excessive growth in the number of tags due to the increase of synonymous tags. Tag redundancy poses a challenging issue in finding the correct tags, forcing users to use their own words for tagging. This issue can also increase the number of duplicate question objects, resulting in more processing delays.

To address the challenges of manual tag recommendation, significant research has focused on developing automatic tag recommendation systems (9). However, this task remains complex, as it needs to consider both the user's varying knowledge and terminology, and the potential for ambiguity in the language. An effective automatic tag recommendation system should still be able to recommend relevant tags that capture the core concept of the question, even when it includes additional details like examples, code snippets, or images.

This study observes that each tag is typically described by a specific combination of words, termed tokens. Consequently, identifying these tag description tokens within a new question's content allows differentiation from other tags. In essence, assigning tag "t" to a new question requires that the question's important tokens also hold significance in other training set questions with label "t." A profile-based approach is employed to achieve this. Here, the important tokens of all questions sharing a specific tag are combined to create a single descriptive profile for that tag. Tag assessment is then performed based on these tag profiles. An adapted version of the Term Frequency-Inverse Document Frequency (TF-IDF) measure is introduced for tag recommendation systems. This novel measure operates at the profile level, calculating the similarity between the new question and tag profiles based on the importance of their terms. Unlike many existing methods, the proposed approach leverages only the raw text of the question object, eliminating the need for text mining tools. This makes it language-independent.

The proposed approach's performance has been evaluated using various datasets from Stack Exchange communities. To further demonstrate language-independence, a novel dataset named PerTagSet was created for Tag Recommendation on Persian CQA systems. This dataset is the first of its kind. A comparison between the proposed approach and a vector space representation baseline yielded significant improvements in experimental results. Additionally, the model's efficiency has been benchmarked against several well-known models in this field, demonstrating superior efficiency. These results highlight the simplicity, higher efficiency, and language-independence of the proposed tag recommendation approach, making it an excellent choice for integration within CQA systems.

Play sound through two or more outputs/devices



147

I have one sound card and one pair of Bluetooth headphones. I want to play my audio through both my sound card and my Bluetooth headphones.



73

I believe Windows has checkboxes that allow you to "check" outputs to enable/disable them, but Ubuntu seemingly has the equivalent of radio selectors (you can only select one at a time).



Bonus question: On a similar note, I have 5 analog output channels on my sound card (in addition to my digital & HDMI audio) -- I would like to be able to determine what comes out of each of those ports (e.g. "front speakers" on all 5 or "front", "center", "back", etc).

sound pulseaudio audio-jack

Figure 1. An example of a question object (from Stack Overflow)

In summary:

- The proposed approach is not dependent on the structure of the information site and determines the suggested tags solely based on the raw text of the question object.
- The proposed approach transcends language barriers. It operates independently of the question's language, eliminating the requirement for language-specific tools like part-of-speech tagging or stemming during data pre-processing. This broadens its applicability to multilingual information sites and simplifies deployment across diverse user bases.
- Our approach utilizes pre-constructed tag profiles, prepared offline. This eliminates the need for real-time comparisons between the input question and a vast collection of sample questions, as employed by many other models.
- The proposed approach achieves high recall rates in both languages compared to baselines and several state-of-the-art models.

The rest of the article is organized as follows: Section 2 is devoted to related works in Question Answering Tag Recommendation. The proposed approach is fully presented in section 3. Sections 4 and 5, respectively, describe the evaluation framework and results. Finally, sections 6 and 7 present the paper's threat to validity and conclusion and offers suggestions for future work.

2. RELATED WORKS

In a Community Question Answering (CQA) system, the Tag Recommendation Module automatically

recommends potential candidate tags to users for their questions. Users can then select several of these recommended tags. Most systems allow users to choose between 1 and 5 tags per question. Several models have been proposed for tag recommendation.

Nguyễn et al. (10) proposed a fuzzy set theory-based technique to recommend tags on IBM Jazz data. Their model also considered user feedback on tags to address software evolution. Xia et al. (11) presented TagCombine, one of the best algorithms for this task. It incorporates three components: multilabel ranking (treating the task as a multi-label learning problem), similarity-based ranking (using tags of similar objects to find candidate tags), and tag-term-based ranking (using the relationship between terms and tags).

Later, Wang et al. (12) introduced EnTagRec, a model implementing two independent modules—Bayesian and Frequentist inference techniques. It combines the results from both probability distributions to determine suggested tags. An enhanced version, EnTagRec++ (12), utilizes user information and the initial set of tags prepared by users to suggest tags.

Li et al. (13) looked at the tag recommendation problem as a multi-label learning problem. They used keyword extraction to strengthen the model and locality-sensitive hashing to improve time complexity. Their proposed approach, MATAR, was evaluated on two different datasets. The use of generative models was also used to determine candidate tags. Wu et al. (14) presented the Tag2Word model that uses tag-word distribution and tag distribution.

Zhou et al. (15) presented the TagMulRec model, which is a collaborative filtering-based tag recommendation model. The model indexes the entire set of objects and searches for objects that are similar to the current question based on the occurrence of words. The model then determines candidate tags using the tag set of the most similar objects. The complexity of the model is linear since the search is done based on the index. Gonzalez et al. (16) considered tag recommendation as a multi-class problem. They used five different classifiers, each to predict one of the tags.

Hong et al. (17) proposed an approach relying on document similarity and historical tag recommendations. To determine document-topic distribution, they filtered out documents with the highest (equal) topic distribution. Next, they calculated the similarity between the input document and filtered documents using the Jensen-Shannon divergence criterion to identify similar documents. Finally, a ranking technique based on the tags of similar documents was used to assign tags.

Maity et al. (18) presented the DeepTagRec, which is a content-cum-user-based deep learning model. The model takes the textual content of the questions as well as the information related to the relationship between users and tags in Stack Overflow. Their evaluation results

Tag Synonyms

 Filter	
Synonym ▾	→ Target ⇅
xubuntu-desktop	xubuntu × 5192
x64	64-bit × 1959
swapfile	swap × 1321
resolution × 933	display-resolution × 1880
cli	command-line × 21481
apt-get	apt × 16319

Figure 2. A small part of the tag synonym page of Ask Ubuntu

on Stack Overflow data show that the model is more efficient than the TagCombine model. However, the model is dependent on the user-tag network, so it may not be applicable to other Q&A systems.

TAGDC (19) is another model that uses machine learning algorithms and collaborative filtering techniques to understand a post. The model consists of two modules, each of which estimates the probability of each tag. The probability results of these two modules are combined, and the top K possible tags are selected from the list of tags.

Recent research also investigates the effectiveness of pre-trained language models for tag recommendation. PTM4Tag (20), utilizes a three-tier architecture that models each post part (title, description, and code) separately and then combines the results. Interestingly, their evaluation suggests that the title of each post provides the most informative content for tag recommendation, potentially aligning with this study's findings regarding the effectiveness of titles for profile identifier generation. This highlights the importance of concise and focused information in capturing the core subject matter for tag suggestion tasks.

Di Rocco and colleagues (21) presented a system called Hybridrec for tagging GitHub repositories. The system uses stochastic and collaborative filtering techniques to find relevant topics. Their model uses Complement Naive Bayes Network to deal with the data imbalance problem. MF-TagRec (22) is another model proposed for tagging GitHub projects. The model employs topics and global semantics of Readme documents and dependency package tags as word vector features.

The tag recommendation task is also applicable to social networks. Banbhrani et al. (23) presented the SCPO model, which uses the Sine Cosine algorithm. The classifier is trained on the characteristics obtained from the set of keywords. The optimizer seeks to find the best value of the classifier parameters.

The literature review shows that despite the presence of several models for the automatic tag recommendation system, a language-independent model that can be employed for low-resource languages' CQA systems has not yet been presented. This is important because the existing models may not necessarily produce an acceptable result for languages with few resources. For example, for Persian CQA systems, text mining tools and Tag-User Network require improvement as they are currently limited and do not provide much information. Also, several of these models use code information to suggest tags, so they will not be suitable for Q&A systems unrelated to programming.

This study aims to develop a language-independent and structure-independent approach to tag recommendation that not only performs well on the standard CQA systems but also can be applicable to the

general topic and low-resource languages. The proposed approach is described in detail below.

3. PROPOSED APPROACH

This study explores the role of tag recommendations in Q&A systems. Automating the tagging process is crucial in this context, as it facilitates users' search for relevant questions and answers. Specifically, given a new question q , a set of existing questions Q and a set of utilized tags T , the goal is to identify a subset of tags from T (*i. e.* T_q) that accurately describe the q .

This paper proposes a profile-based approach for tag recommendation. The approach first constructs a profile for each tag by combining the tokens (important words) from all question objects associated with that tag. Tag assessment is then performed by leveraging these tag profiles. An adapted version of the Term Frequency-Inverse Document Frequency (TF-IDF) measure is introduced to identify the significant tokens within profiles. This novel measure operates at the profile level, calculating the similarity between the new question object and tag profiles based on the importance of their terms. Notably, the proposed approach performs most of the recommendation process offline. In simpler terms, the set of tag profiles is pre-computed. Online processing involves preparing the new question and matching it with existing profiles to identify relevant tags. It is important to highlight that preparing new profiles and questions does not require any natural language processing tools (*e.g.*, part-of-speech tagging and stemming) (24, 25). This characteristic allows the approach to function seamlessly across different languages without modification.

3. 1. Tag Profile Generation

The proposed approach is based on the observation that a specific combination of words typically describes each tag. Therefore, a profile can be generated for each tag to be used in the tag recommendation process. Since the terms in each question's title represent an abstract of that question and contain fewer irrelevant words, these terms have been used to create profiles. To further enhance the accuracy of tag descriptions and the relevance of recommended tags to the input question (minimizing false positives), the study investigates the inclusion of collocations (26, 27) alongside unigrams. In other words, the potential of using bigrams (two-word combinations) to form more representative terms within the profiles is evaluated.

Figure 3 illustrates a title's representative set. Experimental results indicate that using bigrams yields better results than using unigrams (Section 5). Algorithm 1 outlines the steps for generating tag profiles. For each question in the training set, the following steps

Example Title: "finding interlanguage related articles from Wikidump"
Unigram: "finding", "interlanguage", "related", "articles", "from", "Wikidump"
Bigram: "finding interlanguage", "interlanguage related", "related articles", "articles from", "from Wikidump"

Figure 3. An example of n-gram representative words for the title "finding interlanguage related articles from Wikidump"

are taken:

- The n-gram set of the question is extracted (Line 3). The value of n can be either 1 (unigram) or 2 (bigram).
- For each tag associated with the question, the following steps are taken:
 - If the profile already exists in Tag_{prof} , it is fetched (Lines 6-8); otherwise, an empty profile is used (Line 5).
 - The tokens in the n-gram set are added to the profile (Lines 9-11).
- The resulting profile is added to the Tag_{prof} set (Line 12).

3.2. The Proposed Measure This study adapts the Term Frequency-Inverse Document Frequency (TF-IDF) metric to establish a similarity measure for the textual content within recommender systems. First, a description of the original TF-IDF measure is provided, followed by the introduction of the proposed measure. The core principle of the TF-IDF metric lies in representing each document as a vector using weights assigned to its significant words (28-30).

This weighting criterion consists of two terms: Term Frequency (TF) and Inverse Document Frequency (IDF). The first term refers to the occurrence rate of the word w_i in document d_j , while the second term measures how rare the word is in the corpus (i.e. corpus D). Hence, the assigned weight for the word w_i in the document d_j can be calculated as follows:

$$TFIDF(w_{ij}) = tf_{d_j}(w_i) * \log\left(\frac{|D|}{1 + df(w_i)}\right) \quad (1)$$

This equation utilizes $tf(w_{ij})$ to represent the occurrence rate of the i^{th} word (w_i) in document d_j . Important words in document d_j will have a large $tf(w_{ij})$ value. The second term, inverse document frequency (IDF), is defined as the logarithm of the total number of documents in the corpus ($|D|$) divided by the number of documents in the corpus that contain the word w_i ($df(w_i)$). This measure uses the smoothing technique (adding 1 to df) to avoid dividing by zero. An important word should have a small $df(w_i)$ value.

The proposed approach uses the tag profile set to identify relevant tags for the input question. It searches for significant tokens, called representative tokens,

within the profiles of relevant tags that distinguish them from other tags.

To recommend tag t to the input question, the important words in the question must also be important in other questions that have been tagged with t . To effectively quantify the influence of tokens within tag descriptions, this study proposes a novel measure termed Tag Describer Term Frequency-Inverse Collection Frequency. This new measure calculates the similarity between a question object and the profile of a tag. The definition of this metric is as follows:

Algorithm 1 Tag-Profile Generation //This process is offline

Input: Q: set of known question objects with their tags, n: indicator of n-gram

Output: $Tags_{prof}$: set of Tag profiles

```

1.  $Tags_{prof} \leftarrow \{\}$ 
2. for each question  $q$  in  $Q$  do
3.    $ngram_{set} \leftarrow$  extract the set of n-gram collocations from  $q.title$ 
4.   for each tag  $t$  in  $q.tags$  do
5.      $t_{prof} \leftarrow \{\}$ 
6.     if  $t_{prof}$  in  $Tags_{prof}$  then
7.        $t_{prof} \leftarrow Tags_{prof}.get(t_{prof})$ 
8.     end if
9.     for each token in  $ngram_{set}$  do
10.       $t_{prof} \leftarrow t_{prof} \cup token$ 
11.    end for
12.     $Tags_{prof} \leftarrow Tags_{prof}.set(t_{prof})$ 
13.  end for
14. end for
15. return  $Tags_{prof}$ 

```

$$TD-TFICF(w_{ij}) = tf_{d_j}(w_i) * \frac{1}{1 + \sum_{d < j} tf_d(w_i)} \quad (2)$$

where initially, the term frequency of the word w_i within the document d_j is calculated (denoted as $tf_{d_j}(w_i)$). Then, to assess the word's rarity, the inverse of its document frequency (DF) across other documents is incorporated. The one in the denominator is added for smoothing (avoid dividing by zero).

Analysis reveals that frequent words within the document under review, yet rare in other documents, significantly influence tag recommendations. Equation 2 emphasizes this by omitting the logarithmic function from the second part, thereby amplifying the impact of the word's occurrence rate in other documents (collection frequency). Furthermore, instead of solely calculating the document frequency of the word w_i , its frequency across all documents (excluding the document under review) is considered.

Equation 1 traditionally employs the total number of documents within the collection as the numerator in its second term. However, this study adopts a value of 1 in

the numerator. This approach stems from the observation that significant tokens within a new question should also be prominent (frequent) in the tag profiles of its relevant tags. Utilizing the total document count could lead to a decrease in the TF-IDF value. To reduce this problem, the document collection size $|D|$ is excluded from the numerator when calculating the similarity between the new question and candidate tag profiles.

The proposed approach for tag recommendation is described in detail in Algorithm 2. Given a new question q and the tag profile set $Tags_prof$, the algorithm aims to determine k potential tags as suggested tags. The algorithm first extracts the n -grams of question q (line 2). Then, the similarity of the input question and each tag profile is estimated using the similarity measure proposed in this paper (lines 4-9). To do this, the score of each token of the n -grams is calculated based on the frequency of that token in the tag profile (TF) and the total frequency of that token in other documents, i.e., collection frequency (ICF). The value 1 is used to avoid zero division errors. The sum of all tokens in the n -grams of question q yields the degree of similarity of the new question and each tag profile. Finally, the k tags with the highest similarity value are returned as recommended tags for the input question. It should be noted that the TF and ICF of each token can be estimated offline, which contributes to the algorithm's efficiency.

4. EVALUATION FRAMEWORK

This section begins by describing the utilized datasets, including relevant statistics. The PerTagSet dataset, the first Persian tag recommendation dataset, is also introduced. Finally, the evaluation measures commonly employed in tag recommendation literature are presented.

4. 1. Datasets To evaluate the proposed approach, seven different datasets from Stack Exchange communities were used. These datasets, which include

AskDifferent, CordeReview, StackOverflow, DatabaseAdministrator, AskUbuntu, WordPress, and SoftwareEngineering, were prepared by Liu et al. (31) and Gharibi et al. (8). For the cross-language plagiarism detection (32) and a new algorithm based fault tolerance were proposed by Nilchi et al. (33) and Hamidi et al. (34-36). The source of this data can be accessed from Stack Exchange as a data dump in XML format. In the dataset production process, it has been tried to select older questions as well as tags that have been used in at least 50 questions. These tags are more stable because filtered tags are rarely used. Also, the questions for which no tags were left after the above pre-processing were discarded.

Algorithm 2 Profile-based tag recommendation

Input: q : new question, $Tags_prof$: set of Tag profiles, n : indicator of n -gram, k : number of suggest tags

Output: T_q : most probable tags for the new question q

1. $T_q \leftarrow \{ \}$
 2. $ngram_set \leftarrow$ extract the set of n -gram collocations from q .title
 3. $T_score \leftarrow \{ \}$
 4. **for each** t_prof in $Tags_prof$ **do**
 5. **for each** token in $ngram_set$ **do**
 6. $t_score \leftarrow \frac{tf_{t_prof}(token)}{1 + \sum_{doc \in Tags_prof} tf_{doc}(token)}$
 7. **end for**
 8. $T_score \leftarrow T_score \cup t_score$
 9. **end for**
 10. $T_q \leftarrow \text{argmax}^k \text{sort}(T_score)$ //extract k tags with highest scores
 11. **return** T_q
-

Table 1 shows the statistics of these datasets, including the number of questions, the total number of tags, the average number of tags per question, and the average number of questions per tag. As can be seen, the average number of tags in the information sites of the Stack Exchange subcategory is 412. Additionally, approximately three tags are considered for each question. The minimum number of tags per question is one, and the maximum is five.

TABLE 1. Statistical Information of Stack Exchange Tag Recommendation Datasets

Information Site	Number of Final Questions	Number of Final Tags	Average Number of Tags Per Question	Average Number of Questions Per Tag
AskDifferent	77503	469	2.69	449.68
CodeReview	39811	302	2.74	362.13
StackOverflow	47759	428	2.18	243.80
DatabaseAdministrator	50687	293	2.51	440.40
AskUbuntu	48017	414	2.25	261.56
WordPress	70491	403	2.33	413.64
SoftwareEngineering	47531	418	2.34	234.55

To demonstrate the applicability and effectiveness of the language-independent approach, a new dataset named PerTagSet was constructed for Persian tag recommendation. Due to the lack of a comprehensive Persian information site comparable to English counterparts, data from various Persian information sources were utilized to create the dataset. This approach emphasizes the raw text content of questions, excluding code, tables, or images. Since the method is agnostic to the information site structure, the use of diverse sources does not influence the evaluation.

The subject of the questions in the dataset includes general, sport, and programming. Like the English dataset, each question's title, body, and tags were extracted. Beautiful Soup was used to extract data and remove HTML tags.

It is worth mentioning that English words are sometimes used in Persian questions to clarify the meaning, and these words are kept. To have more stable and probable tags, tags that have been used several times were considered for this dataset. Also, the questions with no labels left after this processing were discarded. This dataset is available online. The statistical information of the dataset can be seen in Table 2.

4. 2. Evaluation Measures The proposed method produces a ranked list of recommended tags for the input question. To ensure a consistent and valid comparison, the evaluation measures employed for well-known and state-of-the-art models in this task are adopted here. These measures include Recall@k, Precision@k, and F-Measure@k. The parameter k specifies the number of top-ranked tags generated by the algorithm, which can be either 5 or 10. Let S be the set of questions in the test set. For each question $s \in S$, a set T_s contains the correct tags relevant to that question, while the set R_s contains the k recommended tags generated by the algorithm. The criteria for calculating each measure are as follows.

4. 2. 1. Recall@k It measures the fraction of correct tags that are included in the top k recommended tags. To calculate Recall@k, first calculate Recall@ k_s for each question $s \in S$:

$$\text{Recall}@k_s = \frac{|R_s \cap T_s|}{|T_s|} \quad (3)$$

TABLE 2. Statistical Information of PerTagSet, the Persian Tag Recommendation Dataset

Query	Count
Number of Final Questions	3046
Number of Final Tags	289
Average Number of Tags Per Question	2.18
Average Number of Questions Per Tag	22.97

The average Recall@ k_s is then calculated across all questions within the evaluation set:

$$\text{Recall}@k = \frac{\sum_{s \in S} \text{Recall}@k_s}{|S|} \quad (4)$$

4. 2. 2. Precision@k It measures the fraction of correctly recommended tags out of the total number of recommended tags. To calculate Precision@k, first calculate Precision@ k_s for each question $s \in S$:

$$\text{Precision}@k_s = \frac{|R_s \cap T_s|}{k} \quad (5)$$

The average Precision@ k_s is then calculated across all questions within the evaluation set:

$$\text{Precision}@k = \frac{\sum_{s \in S} \text{Precision}@k_s}{|S|} \quad (6)$$

4. 2. 3. F-Measure@k It is calculated by first calculating the harmonic mean of Recall@ k_s and Precision@ k_s , denoted as F-Measure@ k_s , as follows:

$$\text{F-Measure}@k_s = \frac{2 \times \text{Recall}@k_s \times \text{Precision}@k_s}{\text{Recall}@k_s + \text{Precision}@k_s} \quad (7)$$

The final F-Measure@k is then calculated by averaging F-Measure@ k_s over all questions in the evaluation set:

$$\text{F-Measure}@k = \frac{\sum_{s \in S} \text{F-Measure}@k_s}{|S|} \quad (8)$$

The evaluation for the Stack Exchange datasets follows these steps: a random selection of 10,000 questions is used as the test set, while the remaining questions are employed for training. To guarantee validity of the results, this process is repeated ten times, and the average outcomes are reported. For the PerTagSet dataset, the same evaluation approach is adopted, with a random selection of 300 questions used for testing. The performance of the proposed approach on English Datasets is summarized in Table 3.

5. RESULTS AND DISCUSSION

This section delves into the proposed model's performance through a series of tests designed to address the following three research questions:

- Does the proposed model perform better with unigrams or bigrams?
- What is the comparative performance of the proposed approach against state-of-the-art models on the task of content-based tag recommendation?
- Is the proposed model language-independent and effective for low-resource languages?

The seven Stack Exchange subset datasets described in section 4.1 are employed to address the first and second research questions. To investigate the claim of language-independence (addressed in the third question), the PerTagSet dataset (section 4.2) is utilized.

TABLE 3. Performance of the proposed approach on English Datasets with varying n-gram orders

Information Site	Recall@5		Precision@5		F-Measure@5	
	1-gram	2-gram	1-gram	2-gram	1-gram	2-gram
AskDifferent	0.708	0.723	0.379	0.389	0.489	0.495
CodeReview	0.744	0.758	0.357	0.368	0.471	0.484
StackOverflow	0.703	0.720	0.382	0.392	0.487	0.493
DatabaseAdministrator	0.697	0.715	0.338	0.347	0.438	0.447
AskUbuntu	0.686	0.704	0.351	0.365	0.457	0.471
WordPress	0.641	0.658	0.283	0.297	0.371	0.396
SoftwareEngineering	0.611	0.628	0.278	0.304	0.373	0.387

Information Site	Recall@10		Precision@10		F-Measure@10	
	1-gram	2-gram	1-gram	2-gram	1-gram	2-gram
AskDifferent	0.829	0.847	0.218	0.233	0.335	0.354
CodeReview	0.814	0.828	0.208	0.219	0.319	0.335
StackOverflow	0.783	0.804	0.216	0.234	0.336	0.352
DatabaseAdministrator	0.827	0.846	0.209	0.225	0.327	0.341
AskUbuntu	0.779	0.793	0.194	0.216	0.298	0.322
WordPress	0.786	0.802	0.177	0.193	0.279	0.298
SoftwareEngineering	0.719	0.735	0.158	0.178	0.244	0.276

5. 1. Evaluating the Effect of N-Gram Size on the Proposed Approach

The first experiment investigated the impact of the n-gram value on the performance of the proposed model. The model was evaluated on two different sets of top k recommended tags, $k = \{5, 10\}$. The results showed that the model performed better with $n=2$ (bigram) than with $n=1$ (unigram) for both sets of tags. This is because bigrams take into account the relationship between two consecutive words, while unigrams only consider individual words. As a result, bigrams are able to capture more context. Additionally, studies have shown that the efficiency of the method decreases for higher values of n. This is because the probability of a particular n-gram occurring in a new question decreases.

Furthermore, increasing the value of k from 5 to 10 increased recall but decreased precision on all datasets, as expected. The recall is a measure of how many relevant tags were included in the top k recommendations, while precision is a measure of how many of the top k recommendations were actually relevant. As k increases, the recall will increase because more tags will be included in the recommendations. However, the precision will decrease because more irrelevant tags will be included in the recommendations. Finally, the AskDifferent dataset yielded the best F-measure results, prompting us to note its larger number of unique tags relative to other datasets.

5. 2. Evaluating the Performance of the Proposed Approach Compared to State-of-the-Art Models

The next experiments were conducted to determine the performance of the proposed approach compared to state-of-the-art and well-known models in automatic tag recommendation field. Specifically, the performance of the model was compared to four well-known methods in the field of CQA tag recommendation: TagCombine (11), TagMulRec (15), FastTagRec (31), and Gharibi et al. (8). In addition to the popularity of the used approaches, these models were chosen for evaluation because they were evaluated on similar datasets and according to the standard criteria of this field. Tables 4, 5, and 6 show the results of the comparison with other models based on recall, precision, and F-measure criteria for the top 5 and top 10 recommended tags.

As stated above, the proposed approach demonstrates competitive performance compared to TagCombine, TagMulRec, and FastTagRec models on all datasets. However, it achieves slightly lower performance on some datasets when compared to Gharibi's model (8). This difference is likely due to Gharibi's method incorporating additional data like hyperlinks and code, which can potentially improve performance. While Gharibi's method leverages additional data, this can come at the cost of reduced applicability to information sites with limited such data.

In contrast, the proposed approach is intentionally

designed to rely only on raw textual data in order to be structure and language independent, a core goal of this study. Such an approach is particularly beneficial for information sites where meta data used in Gharibi's method might not be as readily available or relevant.

Furthermore, the proposed approach offers simplicity in implementation, does not necessitate specialized knowledge of the structure or language of the content, and can be easily integrated into various CQA systems without requiring special changes.

TABLE 4. Performance of the proposed approach compared to state-of-the-art models in terms of Recall@k

Recall@5					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.641	0.779	0.708	0.689	0.723
CodeReview	0.689	0.849	0.718	0.758	0.758
StackOverflow	0.641	0.810	0.640	0.698	0.720
DatabaseAdministrator	0.634	0.775	0.666	0.692	0.715
AskUbuntu	0.588	0.760	0.603	0.684	0.704
WordPress	0.553	0.723	0.605	0.632	0.658
SoftwareEngineering	0.540	0.685	0.594	0.582	0.628

Recall@10					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.769	0.876	0.827	0.815	0.847
CodeReview	0.798	0.922	0.788	0.820	0.828
StackOverflow	0.762	0.889	0.749	0.774	0.804
DatabaseAdministrator	0.764	0.872	0.778	0.816	0.846
AskUbuntu	0.717	0.856	0.721	0.770	0.793
WordPress	0.688	0.832	0.725	0.765	0.802
SoftwareEngineering	0.669	0.789	0.704	0.708	0.735

TABLE 5. Performance of the proposed approach compared to state-of-the-art models in terms of Precision@k

Precision@5					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.332	0.407	0.372	0.357	0.389
CodeReview	0.359	0.449	0.377	0.398	0.368
StackOverflow	0.268	0.343	0.343	0.386	0.392
DatabaseAdministrator	0.301	0.374	0.313	0.332	0.347
AskUbuntu	0.251	0.330	0.271	0.346	0.365
WordPress	0.242	0.319	0.265	0.278	0.297
SoftwareEngineering	0.237	0.305	0.253	0.252	0.304

Precision@10					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.203	0.233	0.222	0.216	0.233
CodeReview	0.213	0.249	0.211	0.218	0.219
StackOverflow	0.162	0.191	0.205	0.217	0.234
DatabaseAdministrator	0.185	0.215	0.188	0.201	0.225
AskUbuntu	0.156	0.189	0.166	0.198	0.216
WordPress	0.154	0.188	0.163	0.173	0.193
SoftwareEngineering	0.151	0.180	0.153	0.157	0.178

TABLE 6. Performance of the proposed approach compared to state-of-the-art models in terms of F-measure

F-Measure@5					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.419	0.511	0.488	0.471	0.495
CodeReview	0.454	0.566	0.494	0.502	0.484
StackOverflow	0.363	0.462	0.444	0.476	0.493
DatabaseAdministrator	0.391	0.484	0.426	0.449	0.447
AskUbuntu	0.336	0.439	0.374	0.437	0.471
WordPress	0.321	0.422	0.368	0.386	0.396
SoftwareEngineering	0.315	0.403	0.355	0.352	0.387

F-Measure@10					
Information Site	TagCombine	Gharibi	TagMulRec	FastTagRec	Proposed Approach
AskDifferent	0.311	0.357	0.350	0.342	0.354
CodeReview	0.327	0.382	0.333	0.335	0.335
StackOverflow	0.260	0.306	0.310	0.329	0.352
DatabaseAdministrator	0.290	0.335	0.302	0.323	0.341
AskUbuntu	0.249	0.300	0.270	0.303	0.322
WordPress	0.244	0.297	0.266	0.283	0.298
SoftwareEngineering	0.238	0.283	0.252	0.257	0.276

5. 3. Evaluation of the Proposed Approach on Persian Language Data

The proposed approach was evaluated on Persian language data to demonstrate its language-independent nature and applicability to low-resource languages. The approach was applied to the PerTagSet dataset, which was created for the first time for this language. The dataset contains question objects and their corresponding tags. The proposed approach does not require any changes to be made before applying it to this new data. This is because the approach is not dependent on the structure of the information site and does not rely on any text mining tools. Instead, it identifies possible tags by examining raw text data.

The results presented in Table 7 demonstrate that the proposed approach achieves its best performance on this dataset when using an n-gram size of 2 (Bigram). This is consistent with the findings on the English dataset, which suggests that collocating terms is more effective for tag recommendation. This is because a set of frequently co-occurring terms provides a more precise representation of a tag, which can help distinguish it from other tags based on their tag profiles.

Since the PerTagSet dataset was created for the first time for this language, a vector space representation model was used as a baseline. To provide tag recommendation for a new question object, the baseline method compares the vector representation of the input question with the vector representation of each question

object in the training set. It then considers the tags of the most similar questions as the tags of the input question object.

Table 8 summarized the proposed approach's results compared to the baseline defined for PerTagSet dataset's. The proposed method achieved a significantly higher F1 score than the baseline, which indicates its higher efficiency.

6. THREATS TO VALIDITY

This study introduces a language-independent tag recommendation system for Community Question Answering (CQA) platforms. Potential errors in the experimental setup could compromise internal validity.

TABLE 7. Performance of the proposed approach on a Persian dataset with varying n-gram orders

Recall@5		Precision@5		F-Measure@5	
1-gram	2-gram	1-gram	2-gram	1-gram	2-gram
0.727	0.753	0.265	0.282	0.351	0.388

Recall@10		Precision@10		F-Measure@10	
1-gram	2-gram	1-gram	2-gram	1-gram	2-gram
0.791	0.822	0.179	0.184	0.274	0.291

TABLE 8. Performance of the proposed approach on a Persian dataset compared to the baseline

Model	Recall@5	Precision@5	F-Measure@5
Baseline	0.628	0.227	0.307
Proposed Approach	0.753	0.282	0.388
Model	Recall@10	Precision@10	F-Measure@10
Baseline	0.727	0.172	0.243
Proposed Approach	0.822	0.184	0.291

To mitigate this risk, a thorough review of the dataset and the implementation protocol was conducted. While the possibility of error is always present, the employed benchmarking approach provides a strong level of confidence in the results. The evaluation process aligns with those used in related works, which helps to minimize discrepancies. Additionally, by utilizing older questions in the datasets, the study accounts for potential tag corrections made by users over time, thus strengthening the accuracy of the employed tags.

The model's generalizability to various CQA systems with diverse user demographics and domain-specific content is a critical aspect for external validity. This was addressed by conducting experiments across multiple information sites within the Stack Exchange community, as well as on a Persian-language information site. This demonstrates the model's adaptability to different languages and domains.

Construct validity pertains to the appropriateness of the evaluation metrics. Standard metrics such as Recall@k, Precision@k, and F1-Score@k were employed. These metrics are widely recognized and utilized in the evaluation of tag recommendation systems. However, to ensure a comprehensive evaluation, future work plans to explore additional metrics, like R-Precision, to provide a more nuanced understanding of the model's efficacy.

7. CONCLUSION AND FUTURE WORK

This study proposes a novel profile-based approach for tag recommendation in CQA systems. The method leverages the power of raw text data, eliminating the need for complex text mining tools. Unique tag profiles are created by identifying the most important terms within question objects. The similarity between these terms and the terms in each tag is then calculated using an adapted TF-IDF measure. This approach offers several advantages: it's language-independent, making it applicable to any language regardless of its resources; and it is based on raw text data, eliminating the need for specific question object structures. Evaluations on

various datasets demonstrate that the proposed approach achieves superior or on-par performance compared to baseline and state-of-the-art models, highlighting the effectiveness of this method for tag recommendation in CQA systems.

Looking forward, the study plans to incorporate embedding based approach into the tag profiles for further performance enhancement. Additionally, there is a plan to employ a peak-and-plateau strategy to apply post-processing on the output of the recommendation model. Moreover, the aim is to integrate the approach with fault-tolerant frameworks to ensure reliable tag recommendation services even under system failures. Finally, the potential integration of this tag recommendation system within end-to-end information sites is envisioned, fostering a more streamlined user experience.

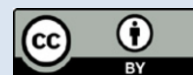
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**Persian Abstract****چکیده**

سیستم‌های پرسش-پاسخ انجمنی در به اشتراک گذاری دانش بسیار مفید هستند. با این حال، با افزایش تعداد پرسش‌ها، مدیریت این سیستم‌ها دشوار می‌شود. پیشنهاد برچسب یک ویژگی حیاتی در سیستم‌های پرسش و پاسخ انجمنی است که به کاربران کمک می‌کند محتوای مرتبط را به راحتی پیدا کنند. متأسفانه، رویکردهای موجود برای توصیه برچسب اغلب وابسته به زبان هستند و یا به ابزارهای پیچیده پردازش زبان طبیعی نیاز دارند، که استفاده از آنها را برای زبان‌های با منابع کم دشوار می‌سازد. برای این منظور، در این پژوهش یک رویکرد مبتنی بر پروفایل برای توصیه برچسب ارائه شده است که مستقل از زبان و ساختار است. رویکرد پیشنهادی بدون نیاز به ابزارهای پیچیده متن کاوی، از داده‌های متنی خام استفاده می‌کند. روش پیشنهادی، بر اساس محتوای سوالات موجود در پیکره، یک پروفایل منحصر به فرد برای هر برچسب ایجاد می‌کند. همچنین یک معیار تشابه اقتباس شده از معیار فراوانی کلمه-معکوس فراوانی سند برای توصیه برچسب‌ها بر اساس پروفایل‌های ایجاد شده ارائه گردیده است. رویکرد پیشنهادی بر روی مجموعه داده‌های مختلف، از جمله زبان انگلیسی و فارسی، ارزیابی شده است. نتایج نشان می‌دهد که این رویکرد به نرخ‌های فراخوان بالایی در مقایسه با مدل‌های پایه‌ای و چندین مدل پیشرفته دست می‌یابد. علاوه بر این، پیاده سازی روش پیشنهادی آسان است و می‌تواند برای زبان‌های با منابع کم استفاده شود. این سیستم می‌تواند به کاربران کمک کند تا محتوای مرتبط را به طور کارآمد در سیستم‌های پرسش و پاسخ انجمنی پیدا کنند.